Design Report of team Bombardilo Crocodil

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This report gives the workflow and process taken up by the team Bombardilo Crocodilo. The problem statement was analysed and suitable configuration of the drone was chosen on the basis of constraints and feasibility. We started with determining basic physical capabilities that the drone must posses to tackle the mission. Then the avionics suite was thoroughly studied and designed. Principles of Aerodynamics and Structure Optimisation were used to define sizing parameters of the drone and finalise the design of the drone and CAD design of the drone was made.

I. Problem Statement

We are assigned to make a logistics drone for a hilly region. It is supposed to replace porters and mules in steep terrain by carrying 10 kg at an altitude of 3,000 m. One way trip is 8km. We have to account for loss of thrust and endurance at an altitude also due to reduced density of air at higher elevations. There is existence of ridge turbulence of upto 12 m/s. We have to make our drone follow terrain for its flight path. Take off weight is under 25 kg .

II. Configuration

A conventional quadcopter allows great accuracy for take off, landing and payload delivery. But such drones have poor range due to low efficiency. Fixed wing drones provide exceptional efficiency and allows great ranges but they require a proper infrastructure (flat surface for landing and takeoff) which is practically impossible at hilly terrains.
To combine best of both worlds, we will use hybrid Quadplane-VTOL configuration. It will takeoff vertically, then fixed wing rotor will start and when enough lift is generated by wing, the VTOL rotors will turn off to maximise efficiency. When drone is at target location, VTOL rotors again kicked in, drone starts hovering and delivery is done via winch mechanism.

III. General Flight Physics

A. VTOL Thrust

The following section outlines general flight physics.

Density of air at sea level

$$\rho_0 = 1.225\,\text{kg/m}^3$$

Density of air at 3000 m

$$\rho_{3000} = 0.909\,\text{kg/m}^3$$

$$\text{reduction} = \frac{\rho_0 - \rho_{3000}}{\rho_0} \times 100 = 25.8\%\ \text{drop}$$

$$\text{T} \propto \rho \quad \text{(VTOL thrust is directly proportional to air density)}$$

$$\text{therefore,} \quad \frac{T_0 - T_{3000}}{T_0} \times 100 = 26\% \quad \text{(For safer side we take 26\%)}$$

To counter turbulence, we specify the thrust-to-weight ratio at 3000 m (TW_{3000}):

$$\text{TW}_{3000} = 2:1 \quad \text{(after reduction in thrust)}$$

Hence, at sea level:

$$\text{TW}_0 = \frac{2}{1 - 0.258} = 2.7:1$$

$$\text{Maximum Takeoff Weight (MTOW)} = 25\,\text{kg}$$

$$\text{Therefore, thrust required} = 2.7 \times 25\,\text{kg}$$

$$= 67.5\,\text{kg}$$

B. Fixed Wing Lift

$$L = \frac{1}{2}\rho V^2 S C_L$$

Lift must be same at sea level and 3000m:

$$\frac{1}{2}\rho_{3000} V_{3000}^2 S C_L = \frac{1}{2}\rho_0 V_0^2 S C_L$$

$$\implies V_{3000} = \sqrt{\frac{\rho_0}{\rho_{3000}}} V_0 = \sqrt{\frac{1.225}{0.909}} V_0$$

$$\implies V_{3000} = 1.16\, V_0$$

Theoretical % increase = 16%

$\therefore$ For safety, cruise airspeed is 17% higher at altitude.

IV. Avionics and Sensor Suite Architecture

A. Flight Controller

Pixhawk 6C

It has built in sensors like gyroscopes, accelerometers, barometers, and magnetometer and can easily handle computation like switching from VTOL mode to fixed wing mode mid flight

B. Firmware

Ardupilot is chosen.

It has built in libraries for UAV and is highly compatible with Pixhawk.

C. Navigation

We use Here4 RTK GNSS module.

Two antenna ensure no error due to intereference with magnets in motors and provide RTK (Real Time Kinematics) and precise heading vector.

D. LiDar Altimeter

Lightware SF11/C

We use this to enable our drone to follow terrain as flight path. Range of upto 100 m gives drone enough time to react to sudden terrain changes.

E. Payload Release Mechanism

- Rhino 60 RPM 12 V High Torque DC Planetary geared motor with integrated optical sensor to provide steady high torque to lift 10 kg without stalling.
- Braided dyneema cable which has high tensile strength and feature practically zero stretch.
- Machined aluminium pulley
- S beam load cell sensor to measure tension in cable. The moment payload is kept at ground, tension falls to 0.
- Electronically controlled mechanical servo release latch releases the payload the moment measure tension falls to 0 and then cable is pulled up to winch motor

All of these electronics can be connected to CAN bus port of flight controller.

F. Telemetry

Holybro Telem long rang radio is a long range telemetry system at 915 MHz and is used for real time contolling and monitoring of battery voltage, flight mode, GPS lock, payload status etc.

V. Power

We need a heavy duty power source that is able to handle huge bursts of currents upto 150 A to 200 A. But at the same time it should be as compact as possible. Hence an energy dense battery pack is a suitable option.
One such power source that can meet the requirements is Li-Po soft case battery pack in 12S1P (12 cells in series, one such pack in parallel). With individual cell voltage of 3.7 V, it can provide voltage of 44.4 V. At full charge the peak voltage can go upto 50.4 V.

A. Key Design Parameters Assumptions

- **Total MTOW (W):** 24.5 kg (Mass $m = 24.5$ kg, Weight force $F = 240.3$ N)
- **Propulsion System:** 4× T-Motor U12 II motors optimized for heavy-lift multirotor/VTOL usage.
- **Target Hover Thrust-to-Weight Ratio (TWR):** 2.0 : 1 (Standard safe threshold for multirotor VTOL control authority).
- **Estimated Hover Power Consumption (P_{hover}):** $\sim$ 200 W per kilogram for a heavy multirotor setup. For 24.5 kg, total electrical hover power is approximately 5000 W (5 kW).
- **Voltage Architecture:** 12S LiPo (44.4 V nominal, 50.4 V fully charged, 39.6 V cutoff).

B. Energy Capacity

Using the nominal voltage of a 12S LiPo pack (44.4 V) and estimated hover power:

$$I_{\text{hover}} = \frac{P_{\text{hover}}}{V_{\text{nominal}}} = \frac{5000\,\text{W}}{44.4\,\text{V}} \approx 112.6\,\text{A}$$

Assuming a design requirement of 10 minutes (0.167 hours) of total vertical VTOL operation (takeoff, landing reserve, and winch payload maneuvering) plus fixed-wing transition buffer:

$$\text{Capacity (Ah)} = I_{\text{hover}} \times \text{Time (hours)} = 112.6\,\text{A} \times 0.167\,\text{h} \approx 18.8\,\text{Ah}$$

In fixed-wing mode, aerodynamic lift from the wings supports the vehicle weight. Assuming a realistic Lift-to-Drag ratio (L/D) of 10 : 1 for a hybrid VTOL airframe:

- **Drag Force (D):**

$$D = \frac{\text{Weight}}{L/D} = \frac{25\,\text{kg} \times 9.81\,\text{m/s}^2}{10} = 24.52\,\text{N}$$

- **Mechanical Thrust Power required at $20\,$m/s (P_{mech}):**

$$P_{\text{mech}} = D \times v = 24.52\,\text{N} \times 20\,\text{m/s} = 490.4\,\text{W}$$

Electrical Cruise Power (P_{cruise}): Accounting for pusher motor and propeller efficiency ($\sim 85\%$ combined):

$$P_{\text{cruise}} = \frac{490.4\,\text{W}}{0.85} \approx 576.9\,\text{W}$$

Current Draw in Cruise (12S ≈ 44.4V):

$$I_{\text{cruise}} = \frac{576.9\,\text{W}}{44.4\,\text{V}} \approx 13\,\text{A}$$

Cruise Capacity (Total $20\,$min $= 0.334\,$h):

$$13\,\text{A} \times 0.334\,\text{h} = 4.235\,\text{Ah}$$

Total Mission Energy Consumed:

$$18.8\,\text{Ah} + 4.235\,\text{Ah} = 23.035\,\text{Ah}$$

To account for a safe 10% discharge reserve (never draining the LiPo below 10% capacity for buffer battery life), we apply a safety factor:

$$\text{Adjusted Capacity} = \frac{23.03\,\text{Ah}}{0.90} \approx 25\,\text{Ah}$$

Therefore, a **12S1P 25000 mAh (25 Ah)** high-discharge LiPo pack is required to sustain safe multirotor and winch operations.

VI. Weight Distribution

We made rough calculation of weight distribution

- Payload = 10 kg
- Battery pack = 3.3 kg
- Airframe and structures (mostly carbon fiber/composite) = 6 kg
- Rotors = 4 kg
- Avionics suite = 1.5 kg

VII. Aerodynamics and Structural Reasoning

A. Wingspan and Chord length Determination

For determining the wing parameters we chose some of these parameters which were realistic for our design:

- **Aspect Ratio of wing =** 7

  A low aspect ratio design helps the drone to safely navigate the specified 12 m/s winds and severe ridge turbulence near steep slopes and makes it easier to keep the drone's structural weight low.

- **Taper ratio of root to chord length =** 0.6

  The wide wing root gives the necessary volume to embed strong internal spars and rigidly bolt the VTOL motor booms. This is critical for absorbing structural vibrations and torque during vertical takeoff.

- **Wing Loading =** $20kg/m^2$

  A higher wing loading make the wing less sensitive to severe wind variations as given in our problem.

- **Zero Swipe wing**

  It helps us to make our aerodynamic center and center of gravity determination simpler and reduce the complexity in determing the internal airframe.

- We use NACA 2412 airfoil for our wing and take ideal 2D lift slope value of $2\pi/rad$.

Based on these specifications we determined other dimension of the wing:

- Ostwald efficiency of wing=0.706
- Area of the wing = $1.25m^2$
- Root chord length of wing =$0.528m$
- Tip chord length of wing =$0.317m$
- Mean chord length of wing =$0.4312m$

- Aerodynamic center of wing = 0.25 of chord length
- Taking stall angle of attack as 14 degrees we get our stall velocity as $15.661 m/s$. Accordingly we take our cruise velocity range from $16 m/s$ to $25 m/s$.

B. Tail sizing

For determining these vertical tail sizing we take the following design assumptions as per our problem requirements:

- **Horizontal tail volume ratio =**0.5

  It provides strong pitch stability and control authority without penalising our structural weight.
- **CG location at 0.3 of chord length from the leading edge.**

  This helps the aircraft to be inherently pitch stable.
- Static Margin of 0.15 to have required stability for the problem.
- **Tail efficiency factor of 0.90.**

  A realistic assumption regarding our design.
- **Vertical tail volume of 0.04.**

  The tail provides aggressive restoring forces to straighten the nose whenever the drone is pushed sideways. This is required for our problem statement where we encounter crosswinds of $12 m/s$.
- **The vertical tail plane is perpendicular to our fuselage axis**.

  For no roll coupling and direct load paths.
- **The vertical tail taper ratio of 0.6.**

  The maximum bending moments caused by violent 12 m/s crosswinds and ridge turbulence occur right at the base connection.
- **A vertical tail arm of** $1.5m$**.**

  It helps us in manually customizing our drone to have enough space to fit the propellers along with satisfying other constraints including our position of neutral point.

Using these assumptions we can get our design parameters as:

- Aspect Ratio of Horizontal tail = 2.923.
- Horizontal Surface area of tail = $0.18 m^2$.
- Tail Horizontal span= $72.47 cm$.
- Tail Horizontal chord= $24.84 cm$.
- Vertical tail span= $34.39 cm$.
- Vertical surface area of tail= $0.0986 m^2$.
- Neutral point at a distance of 0.45 times the length of chord from the leading edge.

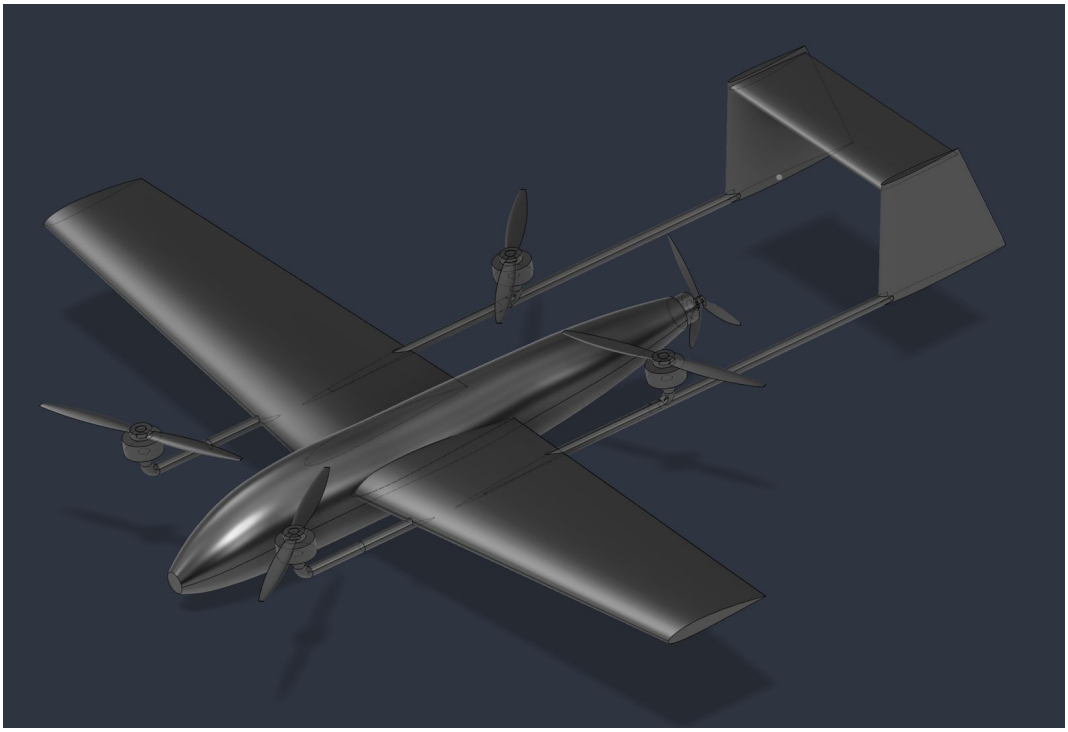


Fig. 1 Full CAD Assembly layout of VTOL Quadplane.

VIII. References

References

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